

Web Services & Service-Oriented Architecture (IT644)

Course placement: Web Services & Service-oriented Architecture (WSSOA) is a core course for the MScIT program in Semester-3.

Course format: 3 hours' lecture and 2-hour laboratory every week.

Course content:

The WSSOA course provides an in-depth exploration of modern distributed computing paradigms, focusing on the design, development, and deployment of interoperable services across heterogeneous platforms. The course covers foundational concepts such as XML, JSON, and RESTful principles, along with advanced topics including SOAP-based services, WSDL, UDDI, and security standards like WS-Security and OAuth. Students will gain hands-on experience with API design, micro-services architectures, cloud-based service deployment, and integration techniques, while critically examining scalability, reliability, and performance considerations in enterprise environments. Through case studies and lab work, learners will bridge theoretical understanding with practical implementation, preparing them for research or industry roles in building robust, scalable, and secure service-oriented solutions.

Books and Literature:

- *Service-Oriented Architecture: Concepts, Technology, and Design* by Thomas E.
- *SOA Principles of Service Design* by T Erl.
- *Service Design Patterns: Fundamental Design Solutions for SOAP/WSDL and RESTful Web Services* by Robert Daigneau
- *Patterns of Enterprise Application Architecture* by Martin Fowler
- *Enterprise Integration Patterns: Designing, Building, and Deploying Messaging Solutions* by Gregor Hohpe & Bobby Woolf
- *Building Microservices* by Sam Newman

Assessment Method: 6-8 lab assignments, a group project, class tests, and end-semester examination.

Course Outcomes:

By the end of the *Web Services and Service-Oriented Architecture* course, postgraduate students will be able to design, implement, and deploy interoperable, scalable, and secure service-based applications that integrate seamlessly across heterogeneous systems. They will demonstrate proficiency in creating both RESTful and SOAP-based services, applying industry standards such as WSDL, UDDI, XML, JSON, WS-Security, and OAuth, and utilizing micro-services and cloud platforms for distributed deployments. Students will be capable of critically evaluating service design patterns, ensuring reliability, scalability, and maintainability in enterprise environments. Furthermore, they will develop the analytical and practical skills to architect service-oriented solutions that address real-world business challenges, bridging the gap between theoretical concepts and professional implementation.

After successful completion of the course the student will have the ability to,

- Understand, analyze, plan, manage, and implement heterogeneous software systems (P1, P2, P12).
- Apply software design principles for cloud implementation / handling software projects using various tools and techniques (P3, P5, P11).
- Plan, monitor, track, and review software projects in a large group setup (P9, P11).
- Create software architecture and related documents, detailed project plans, schedules, along with necessary software implementation strategy. (P10, P11)

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
√	√	√		√				√	√	√	√

Lecture Schedule

Topics / Description	No. of Lectures
Introduction & Foundations	7
Core Web Services Technologies	9
SOA Architecture and Patterns	9
Security, QoS, and Advanced Topics	9
Microservices and Contemporary Trends	6

Introduction & Foundations

- Course overview, objectives, and industry relevance of Web Services & SOA
- Distributed systems and the evolution towards SOA
- Service-orientation principles — abstraction, contract, reusability, autonomy
- Overview of Web service types — REST vs SOAP
- XML and JSON as data exchange formats

Core Web Service Technologies

- SOAP fundamentals — message structure, processing model
- WSDL — service description and structure
- UDDI — service registry and discovery concepts
- REST fundamentals — resources, URIs, statelessness, CRUD mapping
- HTTP methods, status codes, and content negotiation in REST
- Comparative study: SOAP vs REST — use cases and trade-offs

SOA Architecture & Patterns

- SOA reference architecture and components
- Enterprise Service Bus (ESB) — concepts and role in integration
- Service design principles and granularity decisions
- Service composition and orchestration (BPEL, BPMN)
- SOA design patterns — inventory, composition, messaging patterns

- Integration patterns — request/response, publish/subscribe, aggregator, mediator
- Case studies of SOA implementations in enterprise systems

Security, QoS & Advanced Topics

- Security challenges in web services
- WS-Security — SOAP security extensions
- OAuth 2.0, JWT, and API key management for REST services
- Reliability and transactional integrity in services (WS-ReliableMessaging, XA)
- Performance optimization techniques for web services
- Cloud-based SOA — deploying services to AWS/Azure/GCP

Microservices & Contemporary Trends

- Microservices architecture vs SOA — similarities, differences, convergence
- Service discovery, load balancing, and API gateways (Netflix OSS, Kong, NGINX)
- Containerization for service deployment — Docker fundamentals
- Orchestration of services using Kubernetes

Grading components and examination pedagogy

- Lab assessment – 70%
 - Weekly Lab Assignments (20%)
 - Group Project (50%)
 1. Project briefing — problem statement, expectations, deliverables
 2. Project development support — service modeling & integration plan
 3. Mid/Final-project review — testing, debugging, and performance assessment
 4. Project presentations & evaluation
- Class Tests (2) – 10%
- End-Semester Examination (20%)
 - Question types – theory, design, case-study based, application, diagram type.